

AN EXPLICIT OBSTRUCTION TO PRINCIPAL DOMINATION IN THE ERDŐS–STRAUS PAIR SUM

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ABSTRACT. For the complete first-half divisor-pair source we prove that the usual principal-minus-total-nonprincipal-magnitude estimate sums to at most $-p \log p/24$ at every prime $p \equiv 1 \pmod{4}$, $p \geq 2^{20}$. The same conclusion holds when at most J even-character modes are retained per shell once $p \geq 2^{20} J^3$. Exact integer collision bounds and proper even-character observations supply original witnesses without assuming coset saturation. A cubic-quotient example attains the integer bound while all principal-only shell bounds at that prime are nonpositive. No universal ES lower bound is claimed.

1. THE ORIGINAL PAIR SOURCE

Let $p \equiv 1 \pmod{4}$ be prime,

$$I_p = \{a \in \mathbb{Z} : p/4 < a < p/2\}, \quad R = 4a - p, \quad c_a(g) = \#\{d \mid pa : d \equiv g \pmod{R}\}.$$

All residues lie in $G = (\mathbb{Z}/R\mathbb{Z})^\times$. Write

$$m_a = 2\tau_{\text{div}}(a), \quad \mathcal{E}_a = \sum_g c_a(g)^2, \quad T_a = \sum_g c_a(g)c_a(-g), \quad P_a = \frac{m_a^2}{\varphi(R)}.$$

The canonical E/M criteria are $u \mid a^2$ and respectively $R \mid 4u + 1$ or $R \mid 4u + p$. Their normalized coordinates are

$$(h, r, s) = (d_0^2/u, u/d_0, a/d_0), \quad d_0 = \gcd(a, u).$$

The ordered returns are $(a, h\kappa, phr\kappa)$ with $R\kappa = pr + s$, or $(a, phs\lambda, phr\lambda)$ with $R\lambda = r + s$. Clearing denominators verifies $4/p$, and the inverses are $u = a^2/(Ry - pa)$ or $u = pa^2/(Ry - pa)$. These are the classical Type I/II coordinates [1, §2].

The original pair source is $b, c \mid pa$, $R \mid b + c$. Dividing by $\gcd(b, c)$ returns an ordered M pair when the two p -valuations agree; otherwise remove the one remaining p and retain its side to obtain E. If $h = a/(rs)$, the complete fibres are

$$M : (p^\epsilon dr, p^\epsilon ds), \quad d \mid h, \quad \epsilon = 0, 1; \quad E : (pdr, ds) \text{ or } (ds, pdr), \quad d \mid h.$$

Their sizes are $2\tau_{\text{div}}(h)$, so $T_a > 0$ is exactly local occupancy. The full first-half atlas is complete, and ES at p is equivalent to $\sum_{a \in I_p} T_a > 0$; the expanded proof and all fibre maps are supplied with the accompanying workbench source [2].

Orthogonality gives

$$T_a = \frac{1}{\varphi(R)} \sum_{\chi} \chi(-1) \left| \sum_{d \mid pa} \chi(d) \right|^2, \quad \mathcal{E}_a = \frac{1}{\varphi(R)} \sum_{\chi} \left| \sum_{d \mid pa} \chi(d) \right|^2.$$

Thus $T_a \geq L_a := 2P_a - \mathcal{E}_a$. The following theorem addresses that specific estimate, not every possible treatment of the signed sum.

Theorem 1. *For every prime $p \equiv 1 \pmod{4}$, $p \geq 2^{20}$,*

$$\sum_{a \in I_p} L_a \leq -\frac{1}{24} p \log p.$$

Proof. First $\tau_{\text{div}}(n) < 4n^{1/3}$. For primes $q \geq 11$, $e + 1 \leq q^{e/3}$; for $q = 2, 3, 5, 7$ the maxima of $(e + 1)/q^{e/3}$ occur at $e = 3, 2, 1, 1$ and are at most $2, 3/2, 6/5, 11/10$. Successive ratios prove the maxima, and the product of these bounds is $3.96 < 4$.

Also

$$\sum_{\substack{r \leq x \\ r \equiv 3 \pmod{4}}} \frac{1}{\varphi(r)} < \frac{7}{5} \left(1 + \frac{1}{4} \log x\right), \quad x \geq 3.$$

Indeed $r/\varphi(r) = \sum_{d|r} \mu(d)^2/\varphi(d)$. Write $r = dm$; the harmonic sum over the one required odd class of $m \pmod{4}$ is at most $1 + \frac{1}{4} \log(x/d)$ by integration after its first term. Its coefficient sum is

$$\prod_{q \text{ odd prime}} (1 + 1/[q(q-1)]) \leq \exp \left(\sum_{j \geq 1} \frac{1}{2j(2j+1)} \right) = e/2 < 7/5.$$

All series converge absolutely. Consequently, using $a < p/2$ and $2^{-2/3} < 2/3$,

$$\mathcal{P} := \sum_a P_a < \frac{224}{15} p^{2/3} (\log p + 4).$$

The original diagonal gives $\mathcal{E}_a \geq 2\tau_{\text{div}}(a)$. Put $Y = \lfloor \sqrt{p}/2 \rfloor$. Every $d \leq Y$ dividing $a \in I_p$ has a distinct complement above Y , and there are more than $p/(4d) - 1$ such multiples of d . Thus

$$\mathcal{D} := \sum_a \mathcal{E}_a > p \sum_{d \leq Y} \frac{1}{d} - 4Y > p \left(\frac{1}{2} \log p - \log 2 \right) - 2\sqrt{p}.$$

At the stated bound, $p^{1/3} > 100$, $\log p > 40/3$, $\log 2 < 3/4$, and $2/\sqrt{p} \leq 1/512$. Hence

$$\frac{2\mathcal{P}}{p \log p} < \frac{728}{1875}, \quad \frac{\mathcal{D}}{p \log p} > \frac{7}{16}.$$

Since $7/16 - 728/1875 > 1/24$, the result follows. $\square$

For a set Σ_a of at most J even characters, let

$$L_{a, \Sigma_a} = \frac{2}{\varphi(R)} \sum_{\chi \in \Sigma_a} \left| \sum_{d|pa} \chi(d) \right|^2 - \mathcal{E}_a.$$

Orthogonality gives $T_a \geq L_{a, \Sigma_a} \geq 2JP_a - \mathcal{E}_a$. Thus the same summed negative bound holds at $p \geq 2^{20}J^3$. There is an effective all-sublinear extension. For $m \geq 3$, define

$$K_m = \prod_{\substack{q < 2^m \\ q \text{ prime}}} \max_{0 \leq e < 2m} \frac{(e+1)^m}{q^e}, \quad C_m = \min\{C \in \mathbb{Z}_{>0} : C^m \geq K_m\}.$$

The consecutive ratio is below one once $e+1 \geq 2m$, since $(1 + 1/(2m))^m < 2$. Larger primes obey $e+1 \leq 2^e$. Hence $\tau_{\text{div}}(a) \leq C_m a^{1/m}$. If $p \geq 2^{20}$ and $p^{1-2/m} \geq 12JC_m^2$, then

$$\frac{2J\mathcal{P}}{p \log p} < \frac{91}{300}, \quad \boxed{\sum_a L_{a, \Sigma_a} \leq -\frac{1}{8} p \log p.}$$

Indeed $\mathcal{P} \leq (7/5)C_m^2 2^{-2/m} p^{2/m} (\log p + 4)$ and $7/16 - 91/300 > 1/8$. At $m = 4$ the finite product is $K_4 = 127401984/25025$, so $C_4 = 9$ and $p \geq \max\{2^{20}, 944784J^2\}$ suffices. No zero-truncated or integer-refined bound is ruled out by these statements.

2. AN INTEGER BOUND WITHOUT A SATURATION ASSUMPTION

Let $B \leq G$ contain -1 , put $|B| = 2s$, and let $m_C = \sum_{g \in C} c_a(g)$ for each coset. This is a chosen observation subgroup, not an asserted stabilizer of the actual support. For $m = sq + r$, $0 \leq r < s$, set $\mathfrak{b}_s(m) = sq^2 + r(2q+1)$.

Theorem 2.

$$T_a \geq \mathcal{L}_{a, B} := \sum_C \mathfrak{b}_s(m_C) - \mathcal{E}_a \geq L_{a, B} := \frac{2}{|B|} \sum_C m_C^2 - \mathcal{E}_a.$$

The first equality holds precisely when the antipodal-pair totals in every coset differ by at most one. Also $T_a \geq 4 \max(0, \lceil \mathcal{L}_{a, B}/4 \rceil)$.

Proof. Partition each coset into s pairs $\{g, -g\}$ and write $z_j = c_a(g) + c_a(-g)$. Then $T_a + \mathcal{E}_a = \sum_C \sum_j z_j^2$. The least integer sum of squares with fixed total m_C is $\mathfrak{b}_s(m_C)$: moving one unit across a gap of at least two decreases it. Real Cauchy–Schwarz proves the second bound. Finally the commuting involutions $(b, c) \mapsto (c, b)$ and $(b, c) \mapsto (pa/b, pa/c)$ act freely on the original target pairs. The first cannot fix a pair because $R \nmid 2b$; the second would make pa a square; their product would give $bc = pa$ and make -1 a square modulo $R \equiv 3 \pmod{4}$. Therefore T_a is a multiple of four. $\square$

The real bound retains all characters trivial on B , necessarily even, and subtracts the exact remaining collision energy. It does not infer target membership from a coarse occupied coset. If $[G : B] \leq J$, then $L_{a,B} \leq 2JP_a - \mathcal{E}_a$. The proof of Theorem 1 therefore gives

$$\boxed{\sum_a L_{a,B_a} \leq -p \log p / 24 \quad \text{if } p \geq 2^{20} J^3}$$

for arbitrary adaptive per-shell choices of such subgroups. More generally the same proof retains any at most J even-character modes: each squared character sum is at most m_a^2 , and T_a is twice the total even energy minus $\mathcal{E}_a$. This statement does not apply to zero-truncating negative bounds or to the integer refinement.

At the actual prime $p = 87481$, complete enumeration of all 21870 first-half shells gives $L_a \leq 0$ everywhere, even though 34 shells are occupied. The integer refinement with $B = G$ is likewise nonpositive throughout. Thus no nonnegative shell weighting of those bounds certifies this prime.

At $p = 944329$, the principal-only bounds again all fail, but choose

$$a = 236098 = 2 \cdot 97 \cdot 1217, \quad R = 63, \quad B = \langle -1, 13 \rangle,$$

$$B = \{1, 8, 13, 20, 22, 29, 34, 41, 43, 50, 55, 62\}.$$

The original mass is 16, its collision energy is 16, and the index-three coset counts are $(8, 8, 0)$. Hence

$$\mathcal{L}_{a,B} = 12 + 12 - 16 = 8 = T_a, \quad L_a = -16/9, \quad L_{a,B} = 16/3.$$

With $\omega^2 + \omega + 1 = 0$, the retained nonprincipal coefficients are $8(1 + \omega)$ and its conjugate. Each has squared absolute value 64. Both occupied coarse counts become zero modulo two; they retain their original nonempty fibres. An actual pair is $(1, p \cdot 1217)$, returning

$$(h, r, s, \kappa) = (194, 1217, 1, 18242038),$$

$$\frac{4}{944329} = \frac{1}{236098} + \frac{1}{3538955372} + \frac{1}{4067138774169717196}.$$

A separate occupied shell at this same prime has $R = 47$, $T_a = 60$ and negative principal bound. Its group of order 46 has no proper subgroup containing -1 of order greater than two. The terminal order-two observation is the exact target count, not a reduced universal certificate. A proper *spectral* observation does work there. At the original $a = 236094$, take $\chi(5) = \zeta_{23}$ modulo 47, so $\chi(-1) = 1$. Form $C(X)$ by counting original divisors of pa at their discrete logarithms modulo 23. The Laurent norm $Q(X) = C(X)C(X^{-1}) \pmod{X^{23} - 1}$ has

$$Q_0 = 192, \quad (Q_1, \dots, Q_{11}) = (187, 176, 159, 138, 116, 92, 70, 50, 33, 21, 14), \quad Q_{23-j} = Q_j.$$

Exact rational Taylor evaluation gives $1009 < Q(\zeta_{23}) < 1010$ and $24 < Q(\zeta_{23}^2) < 25$. The certificate uses $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$, 18 alternating terms for each arctangent and 18 cosine terms with their next-term bounds; argument uncertainty is bounded using the unit Lipschitz constant of cosine. The expanded proof gives the branch check and a second independent rational enclosure. With collision energy 132 and principal square 2304, retaining only $1, \chi^{\pm 1}, \chi^{\pm 2}$ gives

$$T_a \geq \frac{2304 + 2Q(\zeta_{23}) + 2Q(\zeta_{23}^2)}{23} - 132 > 58, \quad T_a \geq 60.$$

Direct enumeration gives 60. These are five of 23 even modes, not the exact terminal projection. The pair $(2, 327)$ returns $(236094, 780326438241, 4772638766)$. The retained values and collision energy prove the bound; the omitted modes are the projector's explicit kernel coordinates. Their favourable signs have not been assumed.

3. REPRODUCTION AND SCOPE

The standard-library package supplies complete original divisor/pair lists, source factorizations, subgroup tables, inverse marks, rational constants, and two code implementations. The small complete replay covers all 73798 shells at $p \equiv 1 \pmod{4}$ primes through 3000. In the 4519 hard primes through two million, the principal-only failures are exactly 87481, 196561, 944329, 1915201; integer/coset certificates return witnesses at all four. The separate checker exhausts every first-half shell of each failure. No finite census proves a universal inequality. The analytic bound above is proved separately, not inferred from those computations.

The prescribed-prime positivity of $\sum_a T_a$ remains unproved here. The result rules out a specified absolute-value estimate and gives an exact integer replacement retaining useful nonprincipal information. It does not rule out signed arithmetic cancellation, and it makes no historical-priority, Lean-build, or independent-review claim.

REFERENCES

- [1] Elsholtz, C., & Tao, T. (2013). Counting the number of solutions to the Erdős–Straus equation on unit fractions. *Journal of the Australian Mathematical Society*, 94(1), 50–105. <https://arxiv.org/abs/1107.1010v6>. Coordinates: §2, Propositions 2.2 and 2.6.
- [2] The Clankers. (2026, September 19). *Complete-shell collision bounds for the Erdős–Straus equation* [Turn 6 workbench and exact certificates]. Accompanying `core.tex`, `verify.py`, and `check_independent.py`; exact antecedent pair fibres retained from Turn 5, §§3–4.